

第九章 散射问题

§ 9.1 散射问题的一般描述

$$H = H_0 + \hat{U} = \frac{\hat{p}^2}{2m} + \hat{U}, \quad \hat{U} \text{ 为散射势}$$

$$H_0 |\phi\rangle = E |\phi\rangle$$

$$(H_0 + \hat{U}) |\psi\rangle = E |\psi\rangle \quad \text{散射中能量不变.}$$

$$|\psi\rangle = \frac{1}{E - H_0} \hat{U} |\psi\rangle + |\phi\rangle$$

消除分母奇异性: Lippmann-Schwinger 方程.

$$|\psi^\pm\rangle = \frac{1}{E - H_0 \pm i\eta} \hat{U} |\psi^\pm\rangle + |\phi\rangle$$

物理意义:

$$\langle \vec{r} | \psi^\pm \rangle = \langle \vec{r} | \phi \rangle + \int d^3 r' \langle \vec{r} | \frac{1}{E - H_0 \pm i\eta} | \vec{r}' \rangle \langle \vec{r}' | \hat{U} | \psi^\pm \rangle$$

$$\begin{aligned} \delta(\vec{r} - \vec{r}') &= \langle \vec{r}' | E - H_0 \pm i\eta \frac{1}{E - H_0 \pm i\eta} | \vec{r} \rangle \\ &= (E - \frac{\hbar^2}{2m} \nabla^2 \pm i\eta) = \pm i\eta \end{aligned}$$

$$\text{选取 } |\phi\rangle = |\vec{k}\rangle, \quad \hat{K} = \frac{\hat{p}}{\hbar}$$

$$\langle \vec{r} | \phi \rangle = \frac{1}{(2\pi)^{3/2}} e^{i\vec{k} \cdot \vec{r}}$$

$$G^\pm(\vec{r}, \vec{r}') = \langle \vec{r}' | \frac{1}{E - H_0 \pm i\eta} | \vec{r} \rangle$$

$$= \int d^3 k' \langle \vec{r}' | \frac{1}{E - H_0 \pm i\eta} | \vec{k}' \rangle \langle \vec{k}' | \vec{r} \rangle, \quad \langle \vec{r} | \vec{k}' \rangle = \frac{1}{(2\pi)^{3/2}} e^{-i\vec{k}' \cdot \vec{r}}$$

$$= \frac{1}{(2\pi)^3} \int d^3 k' \frac{e^{i\vec{k}' \cdot (\vec{r} - \vec{r}')}}{E - \frac{\hbar^2}{2m} k'^2 \pm i\eta}$$

$$= \frac{1}{(2\pi)^3} \int_0^\infty k'^2 dk' \int_0^{2\pi} d\varphi \int_0^\pi \frac{e^{i\vec{k}' \cdot (\vec{r} - \vec{r}')} \cos\theta}{E - \frac{\hbar^2}{2m} k'^2 \pm i\eta} \sin\theta d\theta$$

$$= \frac{1}{4\pi^2} \frac{1}{|\vec{r} - \vec{r}'|} \int_0^\infty \frac{k' (e^{ik'|\vec{r} - \vec{r}'|} - e^{-ik'|\vec{r} - \vec{r}'|})}{E - \frac{\hbar^2 k'^2}{2m} \pm i\eta} dk' \quad E = \frac{\hbar^2 k^2}{2m}$$

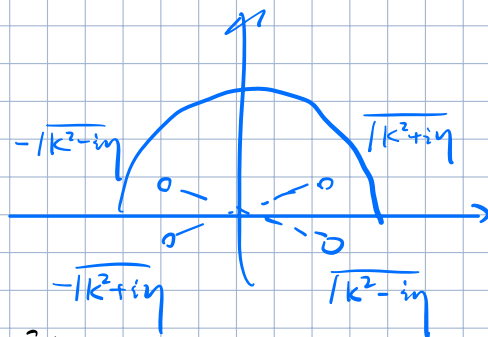
$$= \frac{i}{4\pi^2} \frac{1}{|\vec{r} - \vec{r}'|} \frac{2m}{\hbar^2} \int_{-\infty}^\infty \frac{k' e^{ik'|\vec{r} - \vec{r}'|}}{k'^2 - k^2 \mp i\eta} dk' \quad \int_0^\infty \frac{(-k') e^{-ik'|\vec{r} - \vec{r}'|}}{E - \frac{\hbar^2 k'^2}{2m} \pm i\eta} dk'$$

$$= -\frac{m}{2\pi\hbar^2} \frac{e^{\pm i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} = \int_0^\infty \dots dk'$$

为 $i\eta \frac{2m}{\hbar^2}$, 同阶小 $\rightarrow \eta$ 抵消

$$\text{积分: } \int_{-\infty}^{\infty} \frac{k' e^{i k' |\vec{r} - \vec{r}'|}}{k^2 - k'^2 - i\eta}$$

$$= \oint_{\mathcal{C}} f(k') = 2\pi i \text{Res}[f(\pm k)]$$



另外, 散射势满足 $\langle \vec{r} | \hat{V} | \vec{r}' \rangle = V(\vec{r}) \delta(\vec{r} - \vec{r}')$

$$\langle \vec{r} | \psi^{\pm} \rangle = \langle \vec{r} | \phi \rangle - \frac{m}{2\pi\hbar^2} \int d^3 r' \frac{e^{\pm i k |\vec{r} - \vec{r}'|}}{|\vec{r} - \vec{r}'|} V(\vec{r}') \delta(\vec{r} - \vec{r}')$$

考察渐近行为: $\vec{r} \rightarrow \infty$

$$\left\{ \begin{array}{l} \frac{1}{|\vec{r} - \vec{r}'|} = (\vec{r}^2 + \vec{r}'^2 - 2\vec{r} \cdot \vec{r}')^{-1/2} = \frac{1}{r - \frac{\vec{r} \cdot \vec{r}'}{r}} \\ e^{\pm i k |\vec{r} - \vec{r}'|} = e^{\pm i k r} e^{\mp i \vec{k}' \cdot \vec{r}'} \quad \text{其中 } \vec{k}' = k \frac{\vec{r}'}{r} \end{array} \right.$$

$$\begin{aligned} \langle \vec{r} | \psi^{\pm} \rangle &\xrightarrow{\vec{r} \rightarrow \infty} \langle \vec{r} | \phi \rangle - \frac{m}{2\pi\hbar^2} \frac{e^{\pm i \vec{k} \cdot \vec{r}}}{r} \int d^3 r' e^{\mp i \vec{k}' \cdot \vec{r}'} V(\vec{r}') \langle \vec{r}' | \psi^{\pm} \rangle \\ &= \frac{1}{(2\pi)^{3/2}} \left(\underbrace{e^{\pm i \vec{k} \cdot \vec{r}}}_{\text{入射}} + f(\vec{k}, \vec{k}') \underbrace{\frac{e^{\pm i \vec{k}' \cdot \vec{r}'}}{r}}_{\text{散射}} \right) \end{aligned}$$

$$\text{其中 } f(\vec{k}, \vec{k}') = -\frac{m}{\hbar^2} \int d^3 r' e^{\mp i \vec{k}' \cdot \vec{r}'} V(\vec{r}') \langle \vec{r}' | \psi^{\pm} \rangle$$

$\langle \vec{r} | \psi^+ \rangle$: 出射球面波

$\langle \vec{r} | \psi^- \rangle$: 入射球面波 (含)

$$\text{也可 } f(\vec{k}, \vec{k}') = -\frac{m}{\hbar^2} (2\pi)^{3/2} \langle \vec{k}' | \hat{V} | \vec{k} \rangle \langle \vec{k}' | \psi^+ \rangle \quad \text{傅里叶变换:}$$

$$\int d^3 r' e^{\mp i \vec{k}' \cdot \vec{r}'} \langle \vec{r}' | \psi^+ \rangle = (2\pi)^{3/2} \langle \vec{k}' |$$

$$= -\frac{m}{\hbar^2} \langle \vec{k}' | \hat{V} | \psi^+ \rangle$$

$|\vec{k}| = |\vec{k}'| \rightarrow f(\vec{k}, \vec{k}') \text{ 只与角度有关}$

$$\langle \vec{r} | \psi \rangle \xrightarrow{\vec{r} \rightarrow \infty} \frac{1}{(2\pi)^{3/2}} \left(e^{\pm i \vec{k} \cdot \vec{r}} + f(\theta, \varphi) \frac{e^{\pm i k r}}{r} \right)$$

θ, φ : 以入射粒子动量为轴, 探测器相对于靶的位置。

微分散射截面:

入射流密度。

单位时间被散射到 (θ, φ) 方向之立体角 $d\Omega$ 内的粒子数 $\frac{dn(\theta, \varphi)}{dt} / d\Omega \cdot |\vec{j}_i|$

$$\frac{d\sigma(\theta, \varphi)}{d\Omega} = \frac{dn(\theta, \varphi)}{d\Omega dt} \frac{1}{|\vec{j}_i|}$$

$$\text{总散射截面: } \sigma = \int \frac{d\sigma(\theta, \varphi)}{d\Omega} d\Omega$$

入射粒子流密度. $\vec{J}_i = -\frac{i\hbar}{2m} (e^{-i\vec{k}\cdot\vec{r}} \nabla e^{i\vec{k}\cdot\vec{r}} - e^{i\vec{k}\cdot\vec{r}} \nabla e^{-i\vec{k}\cdot\vec{r}})$

$$= -\frac{i\hbar}{2m} i k [1 - (-1)] = \frac{\hbar k}{m} \quad k = k_r$$

散射粒子流密度. $\vec{J}_s = -\frac{i\hbar}{2m} \left\{ \psi^*(\theta, \varphi) \frac{e^{-i\vec{k}\cdot\vec{r}}}{r} \nabla \left[\psi(\theta, \varphi) \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \right] - \psi(\theta, \varphi) \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \nabla \left[\psi^*(\theta, \varphi) \frac{e^{-i\vec{k}\cdot\vec{r}}}{r} \right] \right\}$

$\frac{dn(\theta, \varphi)}{dt} = \vec{e}_r \cdot \vec{J}_s \quad r^2 d\Omega$ 其中 $\vec{J}_s$ 为散射粒子流密度.

$$\begin{aligned} \vec{e}_r \cdot \vec{J}_s &= -\frac{i\hbar}{2m} \left\{ \psi^*(\theta, \varphi) \frac{e^{-i\vec{k}\cdot\vec{r}}}{r} \frac{\partial}{\partial r} \left[\psi(\theta, \varphi) \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \right] - \psi(\theta, \varphi) \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \frac{\partial}{\partial r} \left[\psi^*(\theta, \varphi) \frac{e^{-i\vec{k}\cdot\vec{r}}}{r} \right] \right\} \\ &= -\frac{i\hbar}{2m} |\psi(\theta, \varphi)|^2 \frac{1}{r} \frac{1}{r^2} [(ikr - 1) - (-ikr - 1)] \\ &= \frac{\hbar k}{m} \frac{|\psi(\theta, \varphi)|^2}{r^2} \end{aligned}$$

$$\frac{dG(\theta, \varphi)}{d\Omega} = \frac{dn(\theta, \varphi)}{d\Omega dt} \frac{1}{|\vec{J}_i|} = \frac{\frac{\hbar k}{m} \frac{|\psi(\theta, \varphi)|^2}{r^2} r^2 d\Omega}{d\Omega \frac{\hbar k}{m}} = |\psi(\theta, \varphi)|^2$$

总散射截面. $G = \int_{\Omega} |\psi(\theta, \varphi)|^2 d\Omega$

§ 9.2 Born 近似

$$\langle \vec{r} | \psi \rangle \xrightarrow{r \rightarrow \infty} \langle \vec{r} | \vec{k} \rangle = \frac{(2\lambda)^{1/2} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' \langle \vec{k}' | \hat{V} | \vec{r}' \rangle \langle \vec{r}' | \psi \rangle$$

$$\begin{aligned} \langle \vec{r} | \psi^\pm \rangle &\xrightarrow{r \rightarrow \infty} \langle \vec{r} | \psi \rangle = \frac{m}{2\lambda\hbar^2} \frac{e^{\pm i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' e^{\mp i\vec{k}'\cdot\vec{r}'} V(r') \langle \vec{r}' | \psi^\pm \rangle \\ &= \langle \vec{r} | \psi \rangle - \frac{m}{2\lambda\hbar^2} \frac{e^{\pm i\vec{k}\cdot\vec{r}}}{r} (2\lambda)^{3/2} \int d^3 r' \langle \vec{k}' | \vec{r}' \rangle V(r') \langle \vec{r}' | \psi^\pm \rangle \\ &= \langle \vec{r} | \psi \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{\pm i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' \langle \vec{k}' | \hat{V}(r') | \vec{r}' \rangle \langle \vec{r}' | \psi^\pm \rangle \end{aligned}$$

迭代求级数解：

$$0 \text{ 阶 } \langle \vec{r} | \psi_0 \rangle = \langle \vec{r} | \vec{k} \rangle, \quad |\psi_0\rangle = |\vec{k}\rangle$$

$$\begin{aligned} 1 \text{ 阶 } \langle \vec{r} | \psi_1 \rangle &= \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \langle \vec{k}' | \hat{V} | \psi_0 \rangle \quad \text{（注：} \vec{k}' = \vec{k} \text{）} \\ &= \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \langle \vec{k}' | \hat{V} | \vec{k} \rangle \end{aligned}$$

$$2 \text{ 阶 } \langle \vec{r} | \psi_2 \rangle = \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \langle \vec{k}' | \hat{V} | \psi_1 \rangle$$

$$= \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' \langle \vec{k}' | \hat{V} | \vec{r}' \rangle \langle \vec{r}' | \psi_1 \rangle$$

$$= \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' \langle \vec{k}' | \hat{V} | \vec{r}' \rangle [\langle \vec{r}' | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}'}}{r'} \langle \vec{k}'' | \hat{V} | \psi_0 \rangle]$$

$$= \langle \vec{r} | \vec{k} \rangle - \frac{\sqrt{2\lambda} m}{\hbar^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \langle \vec{k}' | \hat{V} | \vec{k} \rangle$$

$$+ \frac{2\lambda m^2}{\hbar^4} \frac{e^{i\vec{k}\cdot\vec{r}}}{r} \int d^3 r' \frac{e^{i\vec{k}\cdot\vec{r}'}}{r'} \langle \vec{k}' | \hat{V} | \vec{r}' \rangle \langle \vec{k}'' | \hat{V} | \vec{k} \rangle$$

...

在一阶近似下：注意：\$V(r)\$ 很弱或 \$k\$ 很强。

$$f(\vec{k}', \vec{k}) = - \frac{4\lambda^2 m}{\hbar^2} \langle \vec{k}' | \hat{V} | \vec{k} \rangle$$

$$= - \frac{4\lambda^2 m}{\hbar^2} \frac{1}{(2\lambda)^3} \int d^3 r \langle \vec{k}' | \hat{V}(r) | \vec{r} \rangle \langle \vec{r} | \vec{k} \rangle$$

$$= - \frac{m}{2\lambda\hbar^2} \int d^3 r V(r) e^{i(\vec{k} - \vec{k}')\cdot\vec{r}}$$

\$V(r)\$ 球对称时：\$\vec{q} = \vec{k}' - \vec{k}\$ (\$q = 2k \sin(\frac{\theta}{2})\$) \$\theta\$ 为 \$\vec{k}, \vec{k}'\$ 夹角

$$\frac{d\sigma(\theta, \varphi)}{d\Omega} = |f(\vec{k}', \vec{k})|^2 = \frac{m^2}{4\lambda^2\hbar^4} \left| \int V(r) e^{iqr \cos\theta'} r^2 \sin\theta' dr d\theta' d\varphi' \right|^2$$

$$= \frac{m^2}{\hbar^4} \left| \int_0^\infty V(r) r dr \int_0^\pi e^{iqr \cos\theta'} d(-r \cos\theta') \right|^2 \quad \text{取 } \vec{q} = q \vec{e}_z$$

$$= \frac{m^2}{\hbar^4} \left| \int_0^\infty V(r) r \frac{1}{iq} (e^{iqr} - e^{-iqr}) dr \right|^2$$

$$= \frac{4m^2}{q^2\hbar^4} \left| \int_0^\infty V(r) r \sin qr dr \right|^2 \quad \text{散射分布集中在小角度范围。}$$

§ 9.3 分波法

$V(r)$ 球对称 $\rightarrow$ 角动量守恒

入射波函数 $\rightarrow$ 球谐函数展开 $\rightarrow$ 分别计算.

$$e^{ikz} = \sum_{l=0}^{\infty} i^l \sqrt{4\pi(2l+1)} J_l(kr) Y_{l0}(\theta, \phi)$$

$$\begin{aligned} r \rightarrow \infty &\rightarrow \sum_{l=0}^{\infty} i^l \sqrt{4\pi(2l+1)} \frac{1}{kr} \sin(kr - \frac{l\pi}{2}) Y_{l0}(\theta, \phi) \\ &= \sum_{l=0}^{\infty} i^l \sqrt{4\pi(2l+1)} \frac{1}{2ik} \left[\frac{e^{i(kr - \frac{l\pi}{2})}}{r} - \frac{e^{-i(kr - \frac{l\pi}{2})}}{r} \right] Y_{l0}(\theta, \phi) \\ &= \sum_{l=0}^{\infty} \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi) \frac{1}{2ik} \frac{1}{r} (e^{ikr} - e^{il\pi} e^{-ikr}) \end{aligned}$$

$$\langle r | \psi \rangle \xrightarrow{r \rightarrow \infty} \frac{1}{(2\pi)^{3/2}} \left[e^{ikz} + f(\theta, \phi) \frac{e^{ikr}}{r} \right]$$

$$f(\theta, \phi) = \sum_l \sqrt{4\pi(2l+1)} a_l(k) Y_{l0}(\theta, \phi)$$

$f(\theta, \phi)$ 的投影系数

$$= \frac{1}{(2\pi)^{3/2}} \sum_{l=0}^{\infty} \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi) \left[\frac{1}{2ik} \frac{1}{r} (e^{ikr} - e^{il\pi} e^{-ikr}) + a_l(k) \frac{e^{ikr}}{r} \right]$$

$$= \frac{1}{(2\pi)^{3/2}} \sum_{l=0}^{\infty} \sqrt{4\pi(2l+1)} Y_{l0}(\theta, \phi) \frac{1}{2ikr} \{ [1 + 2ik a_l(k)] e^{ikr} - e^{il\pi} e^{-ikr} \}$$

$$S_l = 1 + 2ik a_l(k) \quad \text{散射出射波.}$$

概率守恒: $|S_l(k)| = 1$

令 $S_l(k) = e^{2i\delta_l}$ δ_l 为相移

$$\begin{aligned} \langle r | \psi \rangle &\xrightarrow{r \rightarrow \infty} \frac{1}{(2\pi)^{3/2}} \sum_l \sqrt{4\pi(2l+1)} \frac{e^{i(\frac{l\pi}{2} + \delta_l)}}{2ikr} (e^{-i\frac{l\pi}{2}} e^{i\delta_l} e^{ikr} - e^{i\frac{l\pi}{2}} e^{-ikr} e^{-i\delta_l}) \\ &= \frac{1}{(2\pi)^{3/2}} \sum_l \sqrt{4\pi(2l+1)} \frac{i^l e^{i\delta_l}}{kr} Y_{l0}(\theta, \phi) \sin(kr + \delta_l - \frac{l\pi}{2}) \end{aligned}$$

从无穷远处看, 弹性散射只改变了出射波的相位.

$$a_l(k) = \frac{e^{i2\delta_l} - 1}{2ik} = \frac{e^{i\delta_l} \sin \delta_l}{k}$$

$$f(\theta, \phi) = \frac{1}{k} \sum_l \sqrt{4\pi(2l+1)} e^{i\delta_l} \sin \delta_l Y_{l0}(\theta, \phi)$$

$$\frac{d\sigma(\theta, \phi)}{d\Omega} = \frac{1}{k^2} \left| \sum_l \sqrt{4\pi(2l+1)} e^{i\delta_l} \sin \delta_l Y_{l0}(\theta, \phi) \right|^2$$

$$\Delta(\theta, \phi) = \int \frac{d\sigma(\theta, \phi)}{d\Omega} d\Omega$$

$$= \frac{4\pi^2}{k^2} \sum_l (2l+1) \sin^2 \delta_l$$

$Y_{l0}(\theta, \phi)$ 的正交性.

$$= \frac{4\pi}{k^2} \operatorname{Im} [f(\theta, \phi)] \quad \text{光学定理}$$

计算相移

径向 $\frac{d^2 u_l}{dr^2} + [k^2 - \frac{2m}{\hbar^2} V - \frac{l(l+1)}{r^2}] u_l = 0 \quad E_k = \frac{\hbar^2 k^2}{2m}$

$r > r_0$ 区域: $u_l(r) = A_l r j_l(kr) + B_l r n_l(kr)$

渐近行为 $j_l(x) \xrightarrow{x \rightarrow \infty} \sin(x - \frac{l\pi}{2}) \cdot \frac{1}{x}$, $n_l(x) \xrightarrow{x \rightarrow \infty} -\frac{\cos(x - \frac{l\pi}{2})}{x}$

$u_l(r) \xrightarrow{r \rightarrow \infty} \frac{1}{k} [A_l \sin(kr - \frac{l\pi}{2}) - B_l \cos(kr - \frac{l\pi}{2})]$

$= C_l \sin(kr - \frac{l\pi}{2} + \delta_l)$

$= D_l r [\cos \delta_l j_l(kr) - \sin \delta_l n_l(kr)]$

利用 $\frac{du_l}{dr} = \frac{d \ln u_l}{dr}$, 确定 δ_l

低能散射: $k \rightarrow 0$, 只考虑 $l=0$ 的 s 分波

在 $r > r_0$ 区域: $\frac{d^2 u_0}{dr^2} + k^2 u_0 = 0 \quad (l=0, r > r_0)$

$\frac{d^2 u_0}{dr^2} = 0$ 解得 $u_0(r) \propto (r - a_0)$

为通解 $u_0(r) \propto \sin(kr + \delta_0)$ 的长波极限。

$\lim_{k \rightarrow 0} \sin(kr + \delta_0) = \lim_{k \rightarrow 0} \sin(kr) \cos \delta_0 + \cos(kr) \sin \delta_0$
 $= kr \cos \delta_0 + \sin \delta_0 \propto r + \frac{\tan \delta_0}{k}$

$\rightarrow$ 散射长度 $a_0 = - \lim_{k \rightarrow 0} \frac{\tan \delta_0}{k}$

微分截面截面:

$f(\theta, \varphi) = e^{i\delta_0} \frac{\sin \delta_0}{k} = \frac{\sin \delta_0}{k(\cos \delta_0 - i \sin \delta_0)} = \frac{1}{k \cot \delta_0 - i k} = -\frac{1}{\frac{1}{a_0} + i k}$

$\sigma(\theta, \varphi) = |f(\theta, \varphi)|^2 = \left| \frac{a_0}{1 + i a_0 k} \right|^2$
 $= \frac{a_0^2}{1 + k^2 a_0^2}$

球形势垒

$$V(r) = \begin{cases} V_0 & r < r_0 \\ 0 & r > r_0 \end{cases} \quad V_0 > 0$$

径向方程:

$$\begin{cases} \frac{d^2 u}{dr^2} - \alpha^2 u(r) = 0, & \alpha = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \\ \frac{d^2 u}{dr^2} + k^2 u(r) = 0, & k = \sqrt{\frac{2mE}{\hbar^2}} \end{cases}$$

边界条件 $u(0) = 0$

$$u(r) = \begin{cases} A \sinh(\alpha r) & r < r_0 \\ B \sin(kr + \delta_0) & r > r_0 \end{cases}$$

连续性条件: $\frac{u'(r_0)}{u(r_0)} = \frac{u'_i(r_0)}{u_i(r_0)}$

$$\frac{\alpha A \cosh(\alpha r_0)}{A \sinh(\alpha r_0)} = \frac{k B \cos(kr_0 + \delta_0)}{B \sin(kr_0 + \delta_0)}$$

$$\begin{aligned} \tan \delta_0 &= \frac{k \tanh(\alpha r_0) - \alpha \tan(kr_0)}{\alpha + k \tanh(\alpha r_0) \tan(kr_0)} \\ &\approx \frac{k [\tanh(\alpha r_0) - \alpha r_0]}{\alpha} \quad k r_0 \ll 1 \end{aligned}$$

$$\begin{aligned} \text{散射长度为 } a_0 &= - \lim_{k \rightarrow 0} \frac{\tan \delta_0}{k} \\ &= r_0 \left[1 - \frac{\tanh(\alpha r_0)}{\alpha r_0} \right] \quad \alpha = \sqrt{\frac{2mV_0}{\hbar^2}} \end{aligned}$$

$V_0 \rightarrow +\infty$ 时, $a_0 = r_0$ 硬球势

球形势阱:

$$V(r) = \begin{cases} -V_0 & (r \leq r_0) \\ 0 & (r > r_0) \end{cases}$$

$$\tan \delta_0 = \frac{k \tan(\alpha r_0) - \alpha \tan(kr_0)}{\alpha + k \tan(kr_0) \tan(\alpha r_0)} \quad \alpha = \sqrt{\frac{2m(E + V_0)}{\hbar^2}} \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\begin{aligned} \text{散射长度 } a_0 &= - \lim_{k \rightarrow 0} \frac{\tan \delta_0}{k} \\ &= - \frac{\tan(\alpha r_0) - \alpha \tan(kr_0)/k}{\alpha + k \tan(kr_0) \tan(\alpha r_0)} \\ &\approx - r_0 \left[\frac{\tan(\alpha r_0)}{\alpha r_0} - 1 \right] \end{aligned}$$

$a_0 < 0$ 有效吸引相互作用
 $a_0 > 0$ 有效排斥相互作用

§ 9.4 全同粒子的散射.

1.

波函数 = 质心波函数 ψ_c + 相对波函数 ψ

交换两粒 $\vec{r} \rightarrow -\vec{r}$, $(r, \theta, \varphi) \rightarrow (-r, \pi - \theta, \pi + \varphi)$

$$\psi(r, \theta, \varphi) \xrightarrow{r \rightarrow +\infty} \propto (e^{i\vec{k} \cdot \vec{r}} \pm e^{-i\vec{k} \cdot \vec{r}}) + [f(\theta, \varphi) \pm f(\pi - \theta, \pi + \varphi)] \frac{e^{ikr}}{r}$$

(1) 玻色子 $S = 0$

$$\frac{dG(\theta, \varphi)}{d\Omega} = |f(\theta, \varphi) + f(\pi - \theta, \pi + \varphi)|^2$$

(2) 费米子 $S = \frac{1}{2}$

同旋单重态,

$$\frac{dG(\theta, \varphi)}{d\Omega} = |f(\theta, \varphi) - f(\pi - \theta, \pi + \varphi)|^2$$

同旋三重态:

$$\frac{dG(\theta, \varphi)}{d\Omega} = |f(\theta, \varphi) - f(\pi - \theta, \pi + \varphi)|^2$$

$\sim$

$\frac{1}{4}$

$\frac{3}{4}$

§ 9.3 非弹性散射

$$\hat{H} = \hat{H}_S + \hat{H}_T + \hat{V} = \hat{H}_0 + \hat{V}$$

粒子: $\hat{H}_S = \sum_m \vec{p}_m^2$ 本征态 $|p\rangle$

靶: $\hat{H}_T |0\rangle = E_0 |0\rangle$

初始状态: $|\psi\rangle = |p\rangle |0\rangle = |p, 0\rangle$

$$\begin{cases} \hat{H}_0 |\psi\rangle = E |\psi\rangle \\ \hat{H} |\psi\rangle = E |\psi\rangle \end{cases}$$

$$|\psi\rangle = |\phi\rangle + \frac{1}{E_0 - \hat{H}_0 \pm i\eta} \hat{V} |\psi\rangle$$

设靶的内部自由度为 z

$$\langle r, z | \psi \rangle = \langle r, z | \phi \rangle + \int d^3r' d^3z' \langle r, z | \frac{1}{E_0 - \hat{H}_0 \pm i\eta} | r', z' \rangle \langle r', z' | \hat{V} | \psi \rangle$$

需要计算格林函数

$$G(r, z; r', z') = \int d^3r'' d^3z'' \langle r, z | \frac{1}{E_0 - \hat{H}_0 \pm i\eta} | r'', z'' \rangle \langle r'', z'' | \hat{V} | \psi \rangle$$

$$= - \frac{m}{2\pi\hbar^2} \sum_z \langle z | 0 \rangle \langle 0 | z' \rangle e^{ikz}$$